

MANOJ KUMAR

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Vijayawada, Andhra Pradesh

Professional Summary

Computer Science graduate who has independently designed and shipped four full-stack applications (Next.js, React, Node.js, PostgreSQL, MongoDB, Prisma, Docker), including a real-time collaborative whiteboard with CRDT-based conflict resolution and an AI coding platform with Docker-sandboxed execution. Solved 150+ Data Structures & Algorithms problems on LeetCode. Experienced in REST API development, JWT authentication, WebSockets, and asynchronous job processing using Redis and BullMQ.

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication, WebSockets

Databases: PostgreSQL, MongoDB, Prisma ORM

Developer Tools: Git, GitHub, Docker, Postman, Vercel, GitHub Actions

Concepts: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, REST Architecture, CI/CD, Agile

Projects

Collaborative Real-Time Canvas

Jun 2026 – Jul 2026

Next.js • React • TypeScript • Node.js • Express.js • PostgreSQL • Prisma • Yjs • WebSockets

[GitHub](#) | [Live](#)

- Engineered a real-time collaborative whiteboard using Yjs CRDTs for conflict-free multi-user editing with live synchronization across concurrent browser sessions.
- Implemented role-based access control (Owner/Editor/Viewer) with invite-by-email, enforcing secure board access through JWT authentication and authorization.
- Architected a scalable Next.js + Prisma + PostgreSQL system supporting auto-save, version history, and comments.

DevPilot AI

May 2026 – Jun 2026

React • TypeScript • Node.js • Express.js • PostgreSQL • Prisma • Docker • Redis • BullMQ

[GitHub](#) | [Live](#)

- Developed an AI-powered coding interview platform covering 100+ problems, executing untrusted user code safely via isolated Docker containers.
- Designed an asynchronous job-queue pipeline with Redis and BullMQ to handle code execution without blocking the main API thread.
- Integrated Monaco Editor with JWT-secured REST APIs for problem submission, execution results, and history tracking.

Zenthra

Apr 2026 – May 2026

Next.js • React • MongoDB • Socket.io • Tailwind CSS

[GitHub](#) | [Live](#)

- Engineered a full-stack social platform with real-time messaging and notifications via Socket.io, backed by secure authentication.
- Architected MongoDB schemas for posts, comments, communities, and user interactions.
- Integrated Cloudinary for media uploads with a responsive Tailwind CSS interface.

AI Resume Intelligence Pro

Mar 2026 – Apr 2026

React • TypeScript • Node.js • MongoDB • JWT

[GitHub](#) | [Live](#)

- Developed a resume intelligence tool that parses PDF resumes, generates ATS compatibility scores, and produces AI-driven improvement feedback.
- Designed REST APIs with Express.js and MongoDB for resume data persistence and retrieval.
- Implemented JWT authentication with per-user history tracking of past resume submissions and scores.

Education

B.E. Computer Science and Engineering

2021–2025

Nehru Institute of Engineering and Technology, Coimbatore

Relevant Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Computer Networks

Certifications

Forage Job Simulations: Datacom (Cloud), Commonwealth Bank (Cybersecurity), Walmart USA (Advanced Software Engineering) – Jul 2026

Achievements

- Solved 150+ Data Structures and Algorithms problems on LeetCode.
- Independently designed, built, and deployed four full-stack applications end-to-end (frontend, backend, database, auth, deployment).